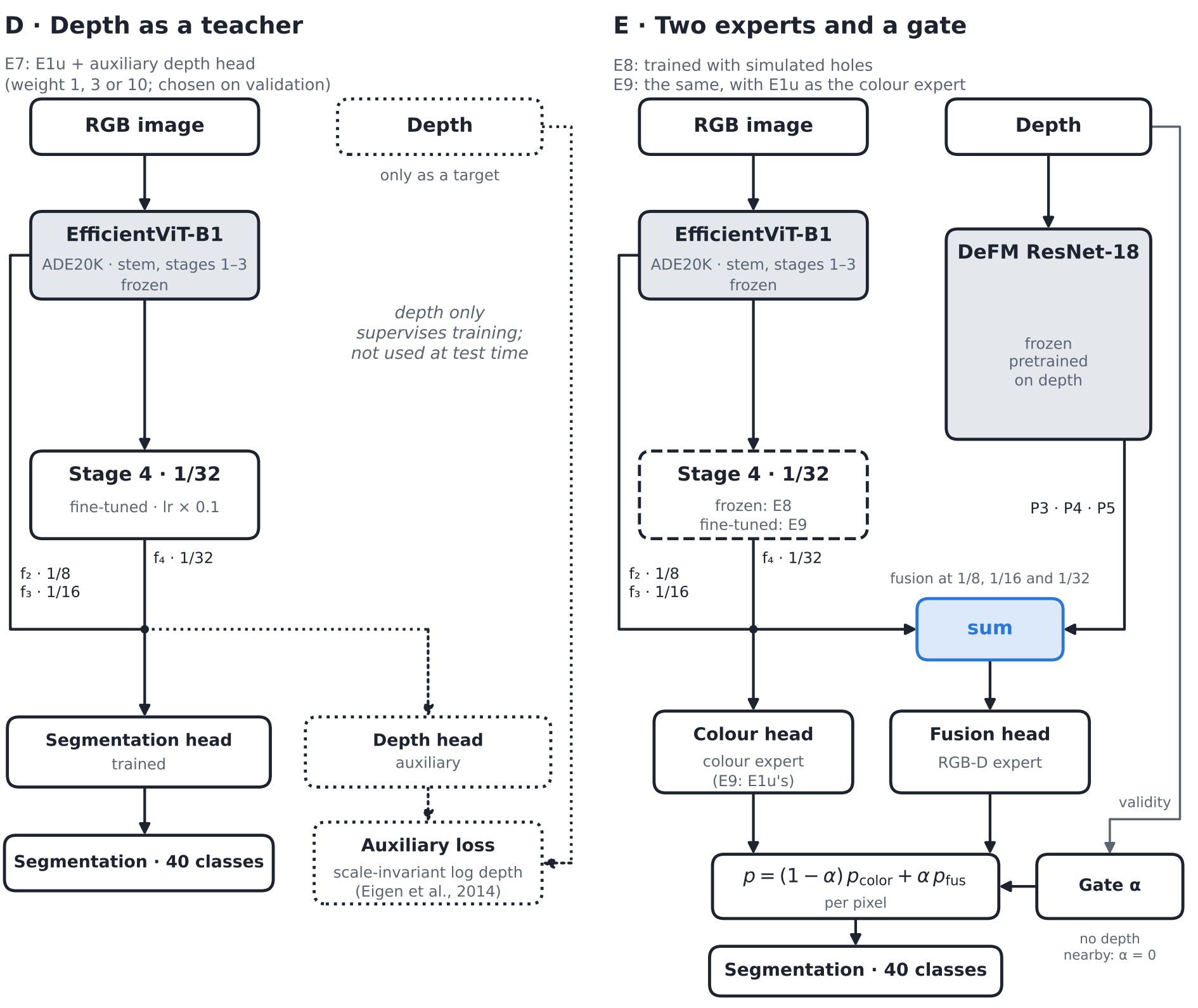
D · Depth as a teacher
E7: E1u + auxiliary depth head
(weight 1, 3 or 10; chosen on validation)
RGB image
Depth
only as a target
EfficientViT-B1
ADE20K · stem, stages 1–3
frozen
depth only
supervises training;
not used at test time
Stage 4 · 1/32
fine-tuned · lr × 0.1
f₄ · 1/32
f₂ · 1/8
f₃ · 1/16
Segmentation head
trained
Depth head
auxiliary
Segmentation · 40 classes
Auxiliary loss
scale-invariant log depth
(Eigen et al., 2014)
E · Two experts and a gate
E8: trained with simulated holes
E9: the same, with E1u as the colour expert
RGB image
Depth
EfficientViT-B1
ADE20K · stem, stages 1–3
frozen
DeFM ResNet-18
frozen
pretrained
on depth
Stage 4 · 1/32
frozen: E8
fine-tuned: E9
P3 · P4 · P5
f₄ · 1/32
f₂ · 1/8
f₃ · 1/16
fusion at 1/8, 1/16 and 1/32
sum
Colour head
colour expert
(E9: E1u's)
Fusion head
RGB-D expert
validity
$p = (1-\alpha)\,p_{\text{color}} + \alpha\,p_{\text{fus}}$
per pixel
Gate α
no depth
nearby: α = 0
Segmentation · 40 classes
frozen
trained
frozen or fine-tuned, depending on the experiment
training only
sum